X-Linked Recessive Ichthyosis

Because the majority of maternal urinary estrogens are derived from the fetal adrenal and are metabolized by the placenta, low levels can reflect fetal abnormalities or death. However, in this condition, low levels do not indicate severe fetal morbidity. The association between failure to initiate or progress labor and ichthyosis in the male offspring was not appreciated until later.

Steroid sulfatase hydrolyzes sulfate esters, including cholesterol sulfate and sulfated steroid hormones. Sulfated fetal adrenal hormones undergo desulfation to form estrogens, which are excreted in maternal urine. The absence of steroid sulfatase enzyme in the fetal-placental unit leads to low maternal urinary estrogen levels and, in some pregnancies, to failure of labor to initiate or progress normally.

In males with **X-linked recessive ichthyosis**, steroid sulfatase enzyme activity is decreased or absent in many tissues, including the epidermis, stratum corneum, leukocytes, and cultured fibroblasts. Cholesterol sulfate, the enzyme's substrate, accumulates in the serum and in skin scales. Carrier females typically have leukocyte steroid sulfatase levels intermediate between those of normal individuals and affected males.

Clinical Features

X-linked recessive ichthyosis occurs in approximately 1 in 2000 to 6000 males. Scaling may begin in the newborn period and is usually most prominent on the extensor surfaces, although significant involvement of flexural areas also occurs.

Although the extent and severity of scaling are variable, X-linked ichthyosis can usually be distinguished clinically from **ichthyosis vulgaris**: - **Ichthyosis vulgaris** is associated with hyperlinear palms and soles, keratosis pilaris, and a family history of atopy. - **X-linked ichthyosis** tends to have more severe scaling with larger, darker scales. Comma-shaped corneal opacities are present in about half of adult patients and do not affect vision. These opacities may also be found in female carriers.

Affected males have an increased risk of **cryptorchidism** and, independently, an increased risk of **testicular cancer**.

Histopathology

Histopathologic examination shows: - Compact orthohyperkeratosis - Acanthosis - Papillomatosis - Thickened granular layer (see Fig 47-2B)

Biochemical and Genetic Findings

Cholesterol sulfate levels are elevated in the serum, epidermis, and scale. On electrophoresis, there is increased mobility of β-lipoproteins (low-density lipoproteins), a feature that may suggest the diagnosis.

Steroid sulfatase is part of a group of arylsulfatases located on chromosome **Xp22**. Over 90% of mutations in X-linked ichthyosis are deletions, which explains the absence of immunologically detectable enzyme protein in some patients. These deletions can often be detected by **fluorescence in situ hybridization (FISH)**, available in many clinical laboratories.

Diagnosis can be confirmed by: - Elevated serum cholesterol sulfate levels (normal: 80–200 μg/dL; X-linked ichthyosis: 2000–9000 μg/dL) - Measurement of arylsulfatase C (steroid sulfatase) activity in cultured fibroblasts (in rare cases without detectable deletions)

Deletions that include adjacent sulfatase genes may lead to **overlap syndromes** involving **chondrodysplasia punctata** and X-linked ichthyosis.

Diagnostic Challenges

Unfortunately, cholesterol sulfate determination in serum and scale may not be readily available in all laboratories for confirmation of the clinical diagnosis.